

Artificial Intelligence

Introduction to Intermediate Level

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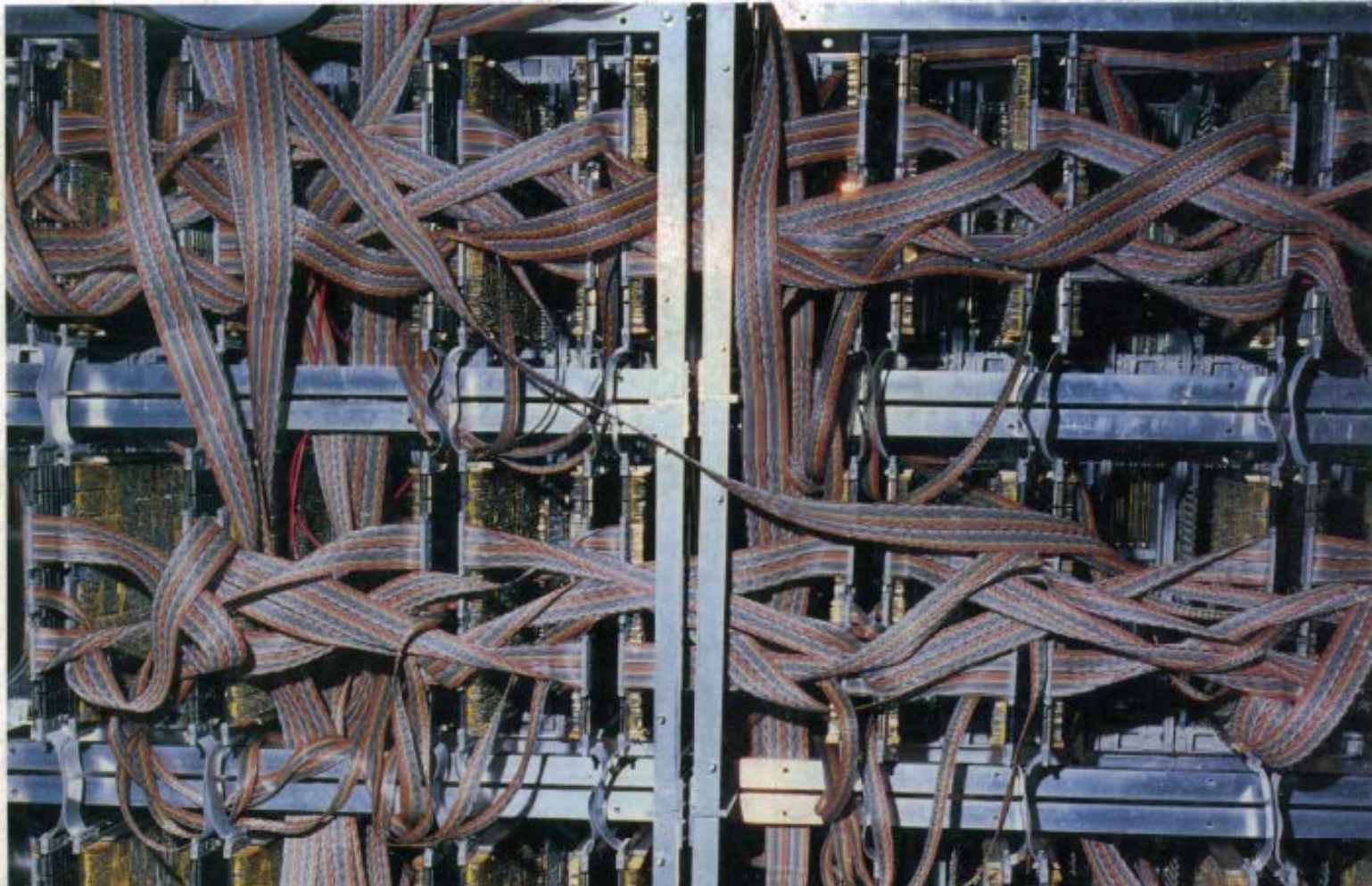
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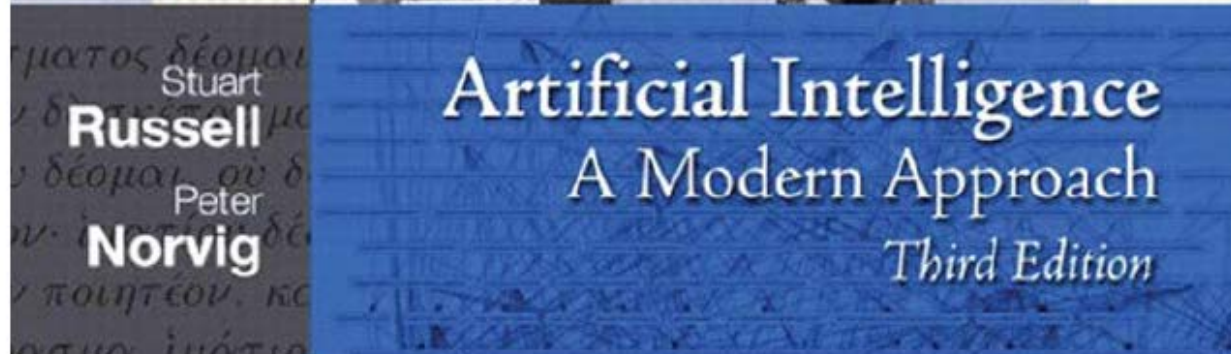
The Columbia Parallel Processor that
I help to built as my Ph.D thesis

18 MARCH 1988

\$3.00

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Textbook

Artificial Intelligence: A modern approach, 3rd Ed

Stewart Russel and Peter Norvig

Artificial Intelligence in the 21 Century

Stephen Lucci and Danny Kopec

Week ↴	Content/ topic/ activity ↴
1 ↴	Introduction. Solving problems by search. BFS search, DFS search. ↴
2 ↴	Uniform-cost search, heuristic search, A* search ↴
3 ↴	Constraint satisfaction problems, origin of NP hard problem ↴
4 ↴	Game play, alpha-beta pruning, Alpha-go ↴
5 ↴	First order logic ↴
6 ↴	Knowledge representation and reasoning using logic, expert systems ↴
7 ↴	Probabilistic Reasoning, Markov chains ↴
8 ↴	Bayes Theorem and applications, Naïve Bayes classification ↴
9 ↴	Bayesian Networks, Midterm exam (cover all materials upto this point) ↴
10 ↴	NLP introduction, NPL tasks, Word2Vec models ↴
11 ↴	NLP word2vec model results and evaluation on NLP downstream tasks ↴
12 ↴	NLP GPT models evaluation ↴
13 ↴	NLP BERT model, evaluation ↴
14 ↴	NPL results presentation and report. ↴

Grading

- Written homework: 24%
- NLP Projects/Computer homework: 30%
- Midterm Exam (cover 2/3 course materials): 40%
- NLP written homework , 6%

Some highly visible recent AI successes in games



Watson defeats
Jeopardy champions
(2011)



DeepMind
achieves human-
level performance
on many Atari
games (2015)



AlphaGo defeats Go
champion (2016)



CMU's Libratus defeats
top human poker
players (2017)

The key technology of game play is **search**.

International chess search space $O(10^{11})$



Go search space $O(10^{170})$



From 1997 IBM Blue to 2016 Google AlphaGo, humans spent 20 years to develop search technology to make

1000000000000000000000000 000000000000000000 (160 zeros)

folds leap to conquer the Go.

in the process, they won Nobel prize

AI Application Areas

- Game Playing
- Navigation
- Autonomous planning and scheduling
- Automated Logistics
- Speech Recognition
- Machine Translation
- Spam Control
- Robotic Vehicles
- Robotics
- Pattern Recognition/Data Mining
- Many other application

What is artificial intelligence?

- Popular conception driven by science fiction
 - Robots good at everything except emotions, empathy, appreciation of art, culture, ...
 - ... until later in the movie.
- Many movies have machines with super intelligence
- But all these are hypes!
- The rice cooker in our home is an AI machine ...
- The invention of computer is far more important than AI ...
- Current AI can not solve many simple problems



Real AI

- A serious science.
- General-purpose (strong) AI
 - Incredibly intelligent, like the robots of science fiction
 - Self learning from experiences
 - Human brain appears to have lots of special and general functions, integrated in some amazing way that we really do not understand
- Special-purpose (weak) AI
 - Solve specific problems
 - E.g., chess/poker/Go playing programs, logistics planning, automated translation, speech and image recognition, web search, data mining, medical diagnosis, keeping a car on the road, ...

Definitions of AI (strong AI perspective)

If our AI system can be **more rational than humans** in some cases, why not?

focus on **action**

sidesteps philosophical issues such as “is the system conscious” etc.

Systems that think like humans	Systems that think rationally
Systems that act like humans	Systems that act rationally

- We will mostly follow “**act rationally**” approach
 - Distinction may not be that important
 - acting rationally/like a human presumably requires (some sort of) thinking rationally/like a human,
 - humans act more rationally in complex situations

- Homework: INTRO1

- “act rationally”

- What does the word “rationally” means?

Hint: In economics the theory assumes players/humans are selfish, but act rationally.

You can use Chinese words here if that's easier.

AI: What is an Agent?

Many big names/concepts

An agent is:

- a rice cooking system/software
- a database system (SQL)
- a driving/control system on your car
- an airline booking system
- a robot in a lab of some routine tasks
- a software system that scan through newspaper articles
-

An agent is a computer/control system

AI is part of Computer Science

Rice cooking system

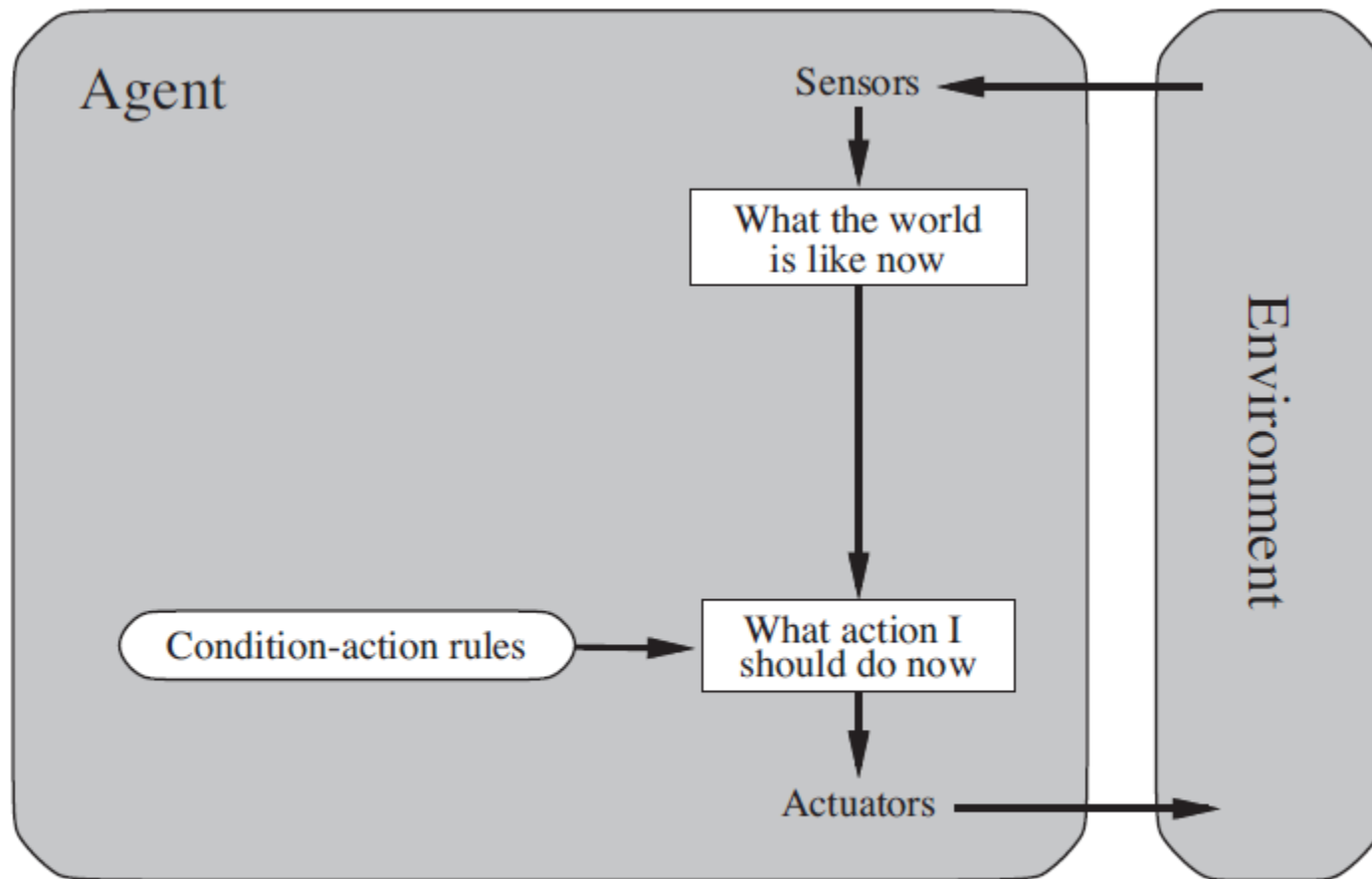


Figure 2.9 Schematic diagram of a simple reflex agent.

Database system

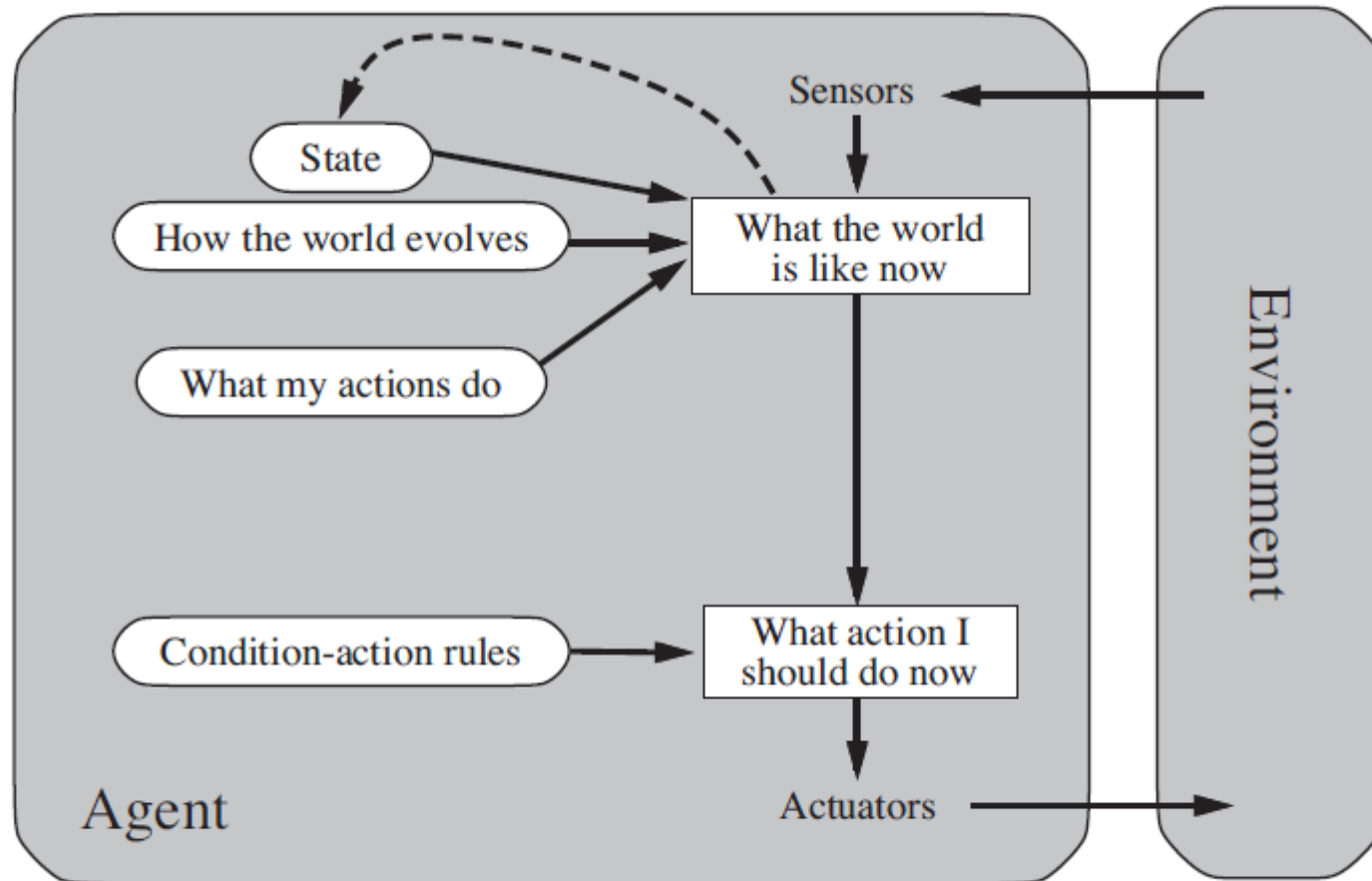


Figure 2.11 A model-based reflex agent.

Car control/driving system

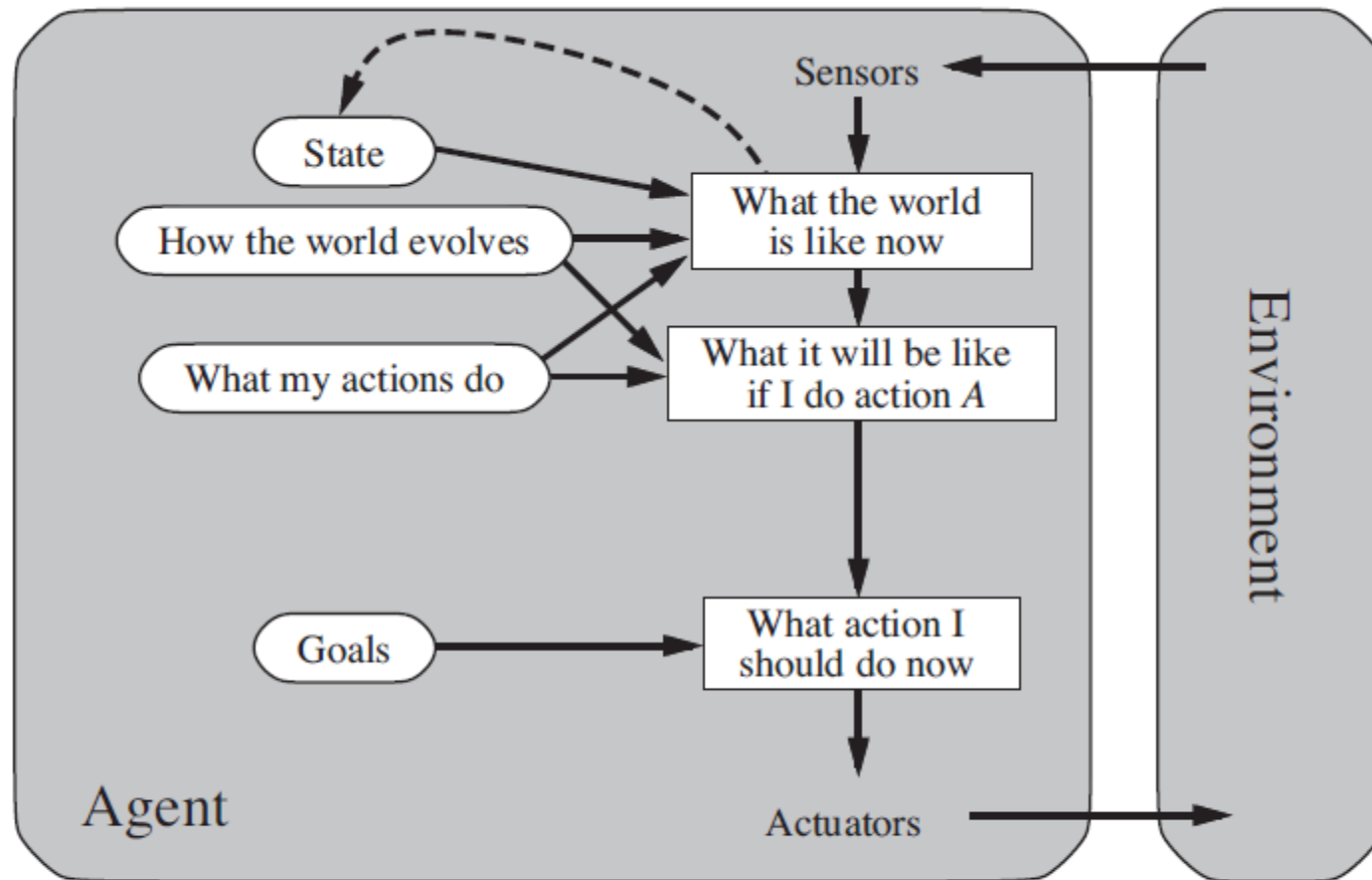


Figure 2.13 A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

A Robot is a very complex AI system



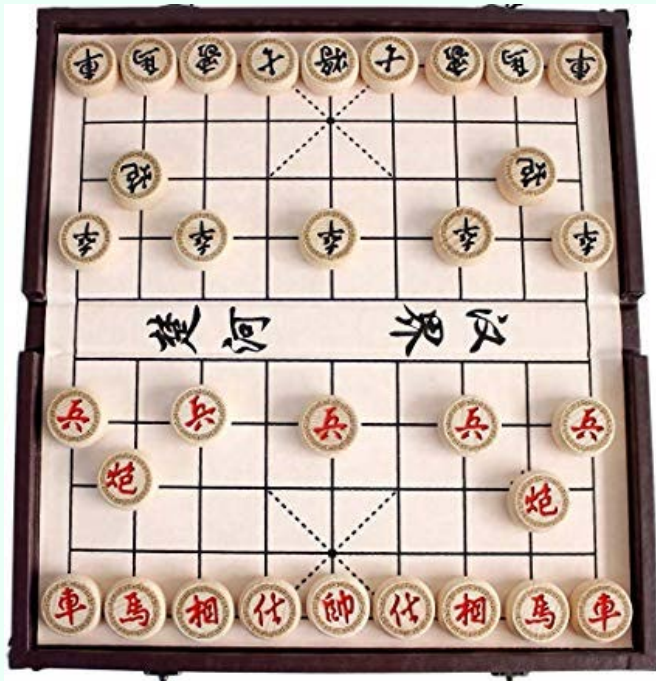
AI techniques and systems

- a rice cooking system/software
- a database system (SQL)
- a driving/control system on your car
- an airline booking system
- a robot in a lab of some routine tasks
- a software system that scan through newspaper articles
-

Many algorithms, techniques, systems used in above computer/control systems

Chess play is the signature activity of human intelligence

**Computer beats World Champion:
Beginning of the age of AI**



A person can see 3 steps ahead.

A computer can see 10 steps ahead, limited by computer memory space.

AI Revolution big success:

1997 Deep Blue defeated World Chess Champion Kasparov



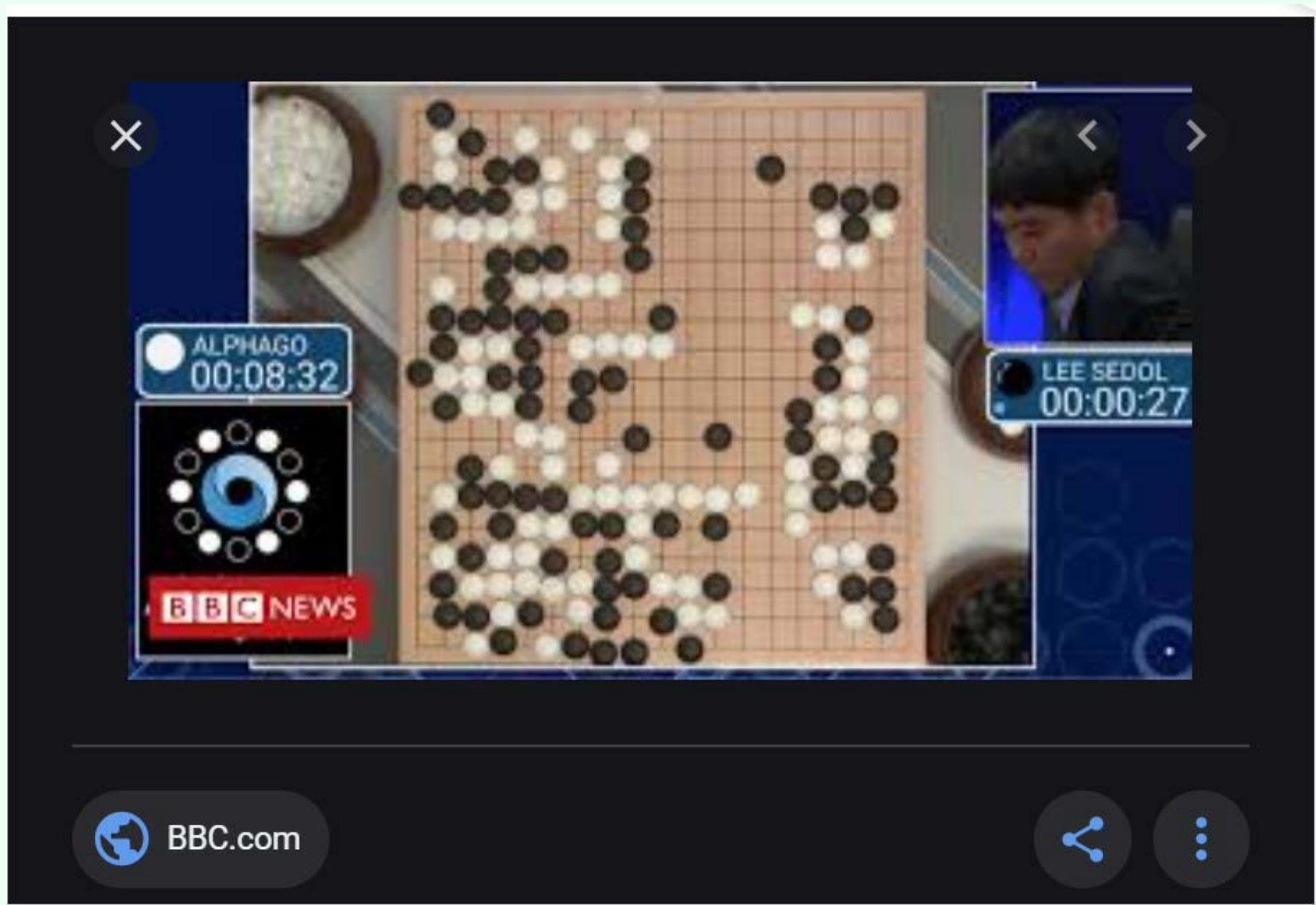
AI Great Achievement by using search related algorithms

1997 Deep Blue defeated World Champion Kasparov



NEW YORK, UNITED STATES: The IBM Deep Blue chess computer team poses in the playing room 08 May 1997 in New York. From L are: Chung-Jen Tan (team manager), Gerry Brody, Joel Benjamin, Murray Campbell, Joseph Hoane and Feng-Hsiung Hsu (seated).

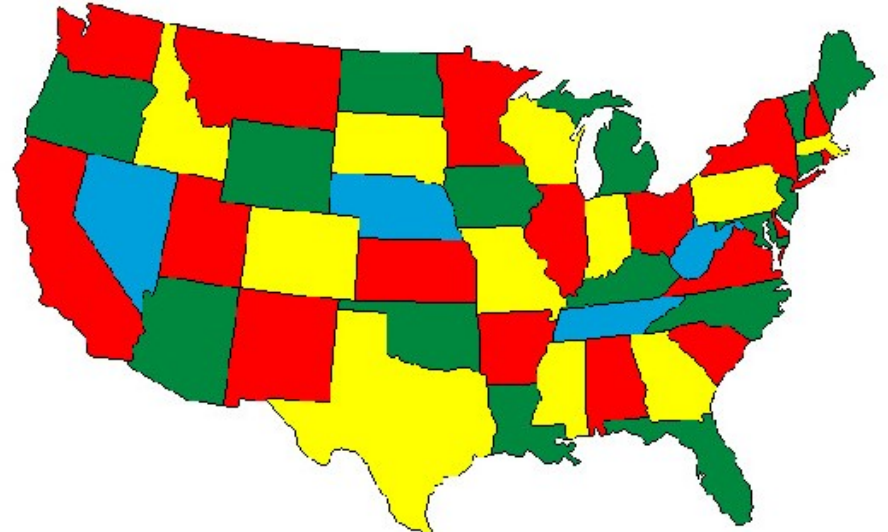
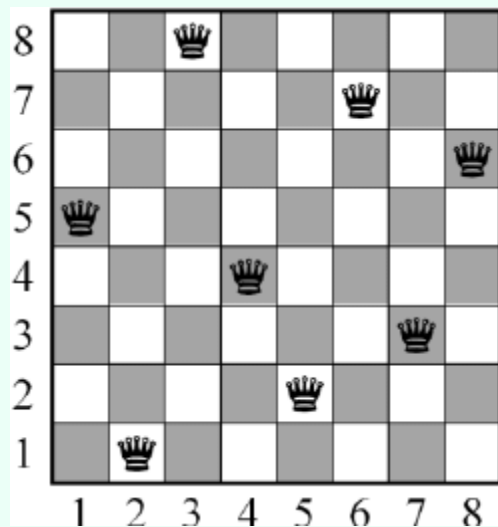
2016 Google AlphaGo beats *World Go Champion* Lee Sedol



Constraint Satisfaction Problem

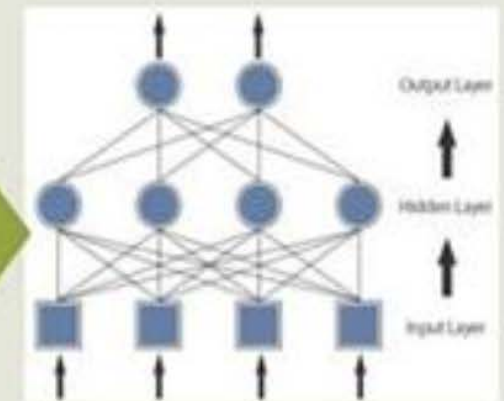
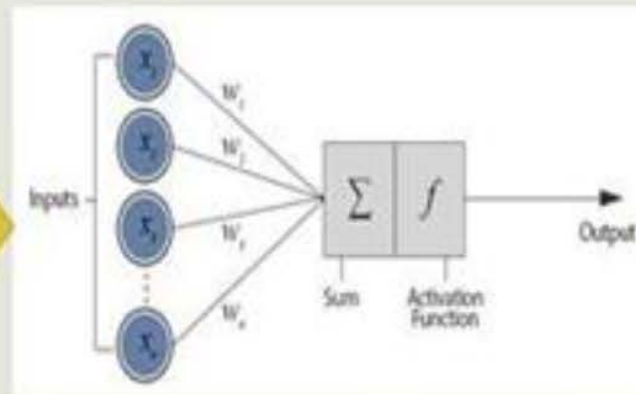
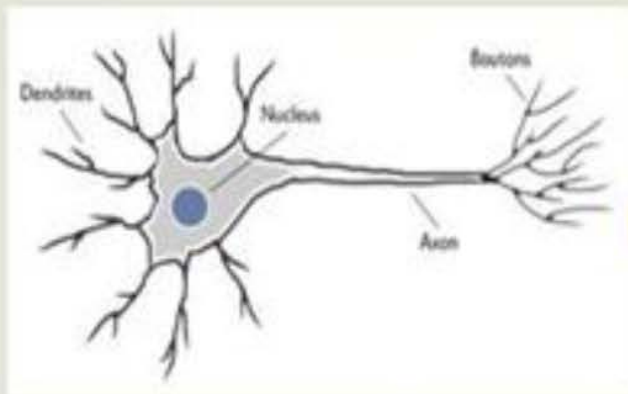


Each puzzle piece is a ***constraint!*** I have to figure out how to do this puzzle!!



Neural Networks

Algorithms that try to mimic the brain. Was very widely used in 80s and early 90s; Currently used as state-of-the-art technique for many Applications



Neuron collects inputs from other neurons using dendrites. The neuron sums all the inputs and if the resulting value is greater than a threshold, it fires. The fired signal is then sent to other connected neurons through the axon.

Neuron connected with n other neurons and thus receives n inputs ($x_1, x_2, \dots, x_n$). This configuration is called a **Perceptron**. There are limitations with this model

Multi-layered neural networks:

Multiclass classification

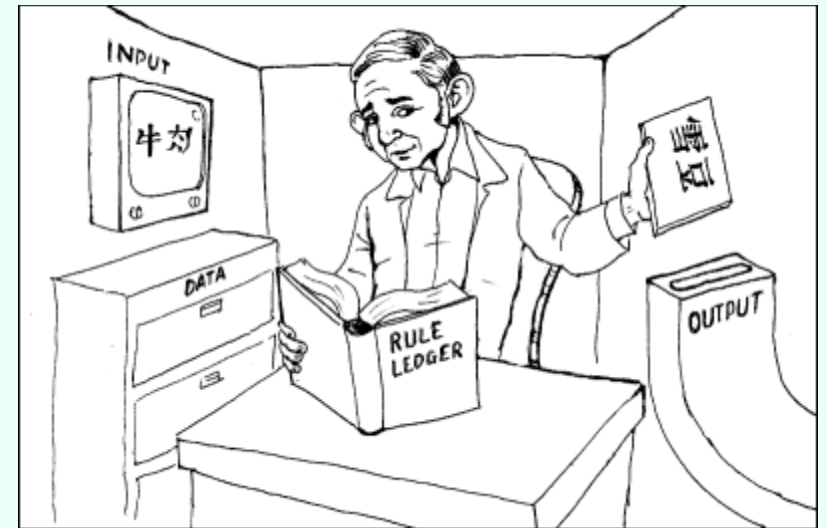
Input Layer: Provides information from the outside world to the network.

Hidden Layer: The hidden layer perform computations thus allowing neural networks to learn more complicated features.

Output Layer: Presents information from the network to the outside world.

“Chinese room” argument [Searle 1980]

image from <http://www.unc.edu/~prinz/pictures/c-room.gif>



- A person who knows English but not Chinese sits in room
- He receives notes in Chinese
- He uses rule book to translate Chinese into English, broadcast out
 - He has no understanding of what they mean
 - But from the outside, it looks like he understand Chinese
- Searle’s argument: the room has no intelligence.
- In fact, all techniques we cover in this course has no intelligence at all.
- Please tell me what intelligence the KNN algorithm has!

Lessons from AI research

- **Clearly-defined** tasks that we think require intelligence and education tend to be doable using AI techniques
 - Playing chess, drawing logical inferences from clearly-stated facts, performing probability calculations in well-defined environments, ...
- **Complex, messy, ambiguous** tasks that come naturally to humans (in some cases other animals) are much harder...
- **Self driving cars** are combination of well defined rules and human errors, a technology very complex --- a busy intersection connecting many roads, cars jam, human crosses, --- accidents cause human life
- **AI systems lack:** broad understanding of the world, common sense, ability to learn from very few examples, truly out-of-the-box creativity...
- **AI don't understand consciousness. ChatGPT has brains?**

Some areas where humans shine

- Coming up with **reasonably good** solutions in complex messy environments
- **Adapting/self-evaluation/creativity** (“My usual approach to chess is getting me into trouble against this person... Why? Is there something entirely different I can do?”)
- **Analogical reasoning, transfer learning** (applying insights from one domain to another)
- **Explaining** our reasoning
- Tasks that require a **broad understanding** of the (human) world
- Knowing **what it's like to be human**
- **Humor**
- ...

Early history of AI

- 1950s/1960s: Early founders met and plan AI. Proved some theorems. Some initial work on neural networks...
- 1970s: Pattern recognition research, to automatically recognize handwritten letters. Proved initial neural network cannot solve exclusive-OR problem.
- 1980s: Hopfield networks re-ignite neural network research, back-propagation. Creation of **expert systems** (systems specialized for one particular task based on experts' knowledge), wide industry adoption

History of AI...

- 1990s: text processing, search engines, data mining research grow rapidly due to computer to collect and generate enormous data
- 1997: computer beats World Champion
- 2000's: machine learning take off. research grow, many algorithms/technologies become mature.
- 2010s: deep learning, pushing computer vision, NLP grow

Modern AI

- More rigorous, scientific, formal/mathematical
- Fewer grandiose promises
- Divided into many subareas interested in particular aspects of human activities, driving, chess play, etc
- More directly connected to “neighboring” disciplines
 - Theoretical computer science, statistics, economics, operations research, biology, psychology/neuroscience, ...
 - Often leads to question “Is this really AI”?
- Some senior AI researchers are calling for re-integration of all these topics, return to more grandiose goals of AI (strong AI)
 - Somewhat risky research propositions for graduate students and junior faculty...

Healthy Perspective of AI

- The purpose of AI is to improve/innovate technology
- Some tasks, humans' way is not necessarily better.
- Some tasks humans don't do well:
 - Compare two pictures of the same person to check if they are some person
- The over-all drive is to develop new/effective technology to solve more tasks
 - Some used to be done by humans
 - Many new tasks, like new city planning, control, etc
- Other technology such as using big data, data driven methods are not necessarily AI.

Homework Intro2: Can you think of a task humans don't do well.

This course covers:

fundamental (early) bread/butter techniques with some applications.

Will not cover machine learning, neural networks, because most/all students take separate specialized courses.

Will do hands-on experiments on NLP.